MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 13 (15-19 Nov 2026) - Period 1	Unit 4 - 4.2 Core Logic of the Game (2/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 13 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 85% correct on range formula; Math.floor reason weaker. Today: scoring, tray, missed worms, results and edge cases.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students complete the game logic: score and worm count, add worms to the tray, calculate missed worms, display results with alert, return to index, use a gameOn variable and fix edge cases. Sprint 4.2 obj. 5-11; CSTA 3A-AP-21.

Success criteria — *what will successful learning look / sound like?*

My score increases only when a visible worm is clicked; missed worms = total - caught; results show at the end; clicks after game over do nothing.

Key knowledge / skill / understanding:

Variables store state; conditions check gameOn; edge cases are unusual inputs that break logic.

Likely misconception / access or challenge need:

Clicking fast counts multiple times is fine; testing once is enough.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 12 min

What students will do:

Code along: score and worm-count variables, click listener on worms, add worm image to tray, missed = total - caught.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Project source; student book 4.2.

Support (low performers / SEN):

Skeleton code; checklist of variables.

Stretch (fast learners / gifted):

Store high score in sessionStorage.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 18 min

What students will do:

Implement results alert, back to index, and gameOn flag; then hunt edge cases (double click, click after end, empty name) and fix at least two.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Edge-case card; browser console.

Support (low performers / SEN):

Edge-case card with hints for 2 cases.

Stretch (fast learners / gifted):

Write a short test plan table with 5 cases and expected results.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Playtest swap: pairs play another pair's game and log bugs in a shared sheet (bug, steps, expected, actual).

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Shared Teams/Excel bug log.

Support (low performers / SEN):

Bug log with one example row.

Stretch (fast learners / gifted):

Prioritise bugs by severity.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: a buggy version where score goes up when clicking empty holes. Model variables for state, the gameOn check, and how a test plan works. Explain playtesting roles.

What is new or innovative in this lesson? *(ETEC looks for this)*

Professional playtesting with a shared bug log; students learn that software is tested by others before release.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Score updating on hidden worms; alert loops; gameOn never reset; students fixing bugs by deleting features. Response: 'it works on my computer' - test differently.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write the formula for missed worms and one edge case you fixed. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: fix one bug from your playtest log; upload before/after code screenshot.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Bug log was a hit; students from different classes in later sections used the same format. 70% have a fully working game loop.

What will I change next time?

Cut the modelling to 10 minutes; give more time to edge cases. Prepare a 'common bugs' slide.

Students needing follow-up:

Students without a working game (approx. 5 per class): provide fixed source to continue; contact parents of 2 students with repeated missing work.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☐C ☐H	☑1 ☐2 ☐3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 13 (15-19 Nov 2026) - Period 2	Unit 4 - 4.3 Enhancements + 4.4 Animations

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 13 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Games with full logic for ~70%; bug logs good. Last lesson before the autumn break: polish the game with a custom cursor, popup, sound and animation.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students enhance the worm game: glove cursor, styled results popup, catch sound with mute, and a CSS @keyframes glove animation triggered from JavaScript. Sprint 4.3-4.4; CSTA 3A-AP-16, 3A-AP-19.

Success criteria — *what will successful learning look / sound like?*

My glove cursor works with a fallback; the results popup replaces alert(); a sound plays on catch and can be muted; the glove animates when catching a worm.

Key knowledge / skill / understanding:

cursor: url() with fallback; classList show/hide; Audio() plays after a click; @keyframes + add/remove class replays an animation.

Likely misconception / access or challenge need:

Any image size works as a cursor; music can autoplay; replaying an animation needs a page refresh.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Discussion / dialogue **Time:** 5 min

What students will do:

Compare a plain and an enhanced version of the game; agree 3 UX rules (clear, not annoying, player in control).

Evidence tool: Oral questioning

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Two demo builds.

Support (low performers / SEN):

Picture comparison with key words.

Stretch (fast learners / gifted):

Name an accessibility risk of sound-only feedback.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 26 min

What students will do:

Code along in 4 short steps: 1) glove cursor; 2) results popup with Restart; 3) catch sound + mute button; 4) @keyframes swing triggered by .catch class removed on animationend.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; QR 'Customize the cursor', 'Create a customized popup', 'Audio elements', 'CSS animations'.

Support (low performers / SEN):

Popup, audio and keyframes CSS ready; students write the JS lines only.

Stretch (fast learners / gifted):

Popup scale-in animation and prefers-reduced-motion support.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Stations / carousel **Time:** 6 min

What students will do:

Quick playtest carousel: play two classmates' games and leave one 'star' and one 'wish' on a sticky note or Teams.

Evidence tool: Peer-assessment checklist

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Sticky notes or Teams post.

Support (low performers / SEN):

Feedback phrases card.

Stretch (fast learners / gifted):

Feedback names the exact feature and fix.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: play the plain game, then the enhanced one. Model each enhancement for about 1 minute: cursor CSS, popup classList toggle with z-index, new Audio().play() inside the click handler, @keyframes with 0/50/100% and animationend. Remind: save a backup first.

What is new or innovative in this lesson? *(ETEC looks for this)*

Four polish features in short modelled steps let every student finish a complete, playable game before the break; peers playtest it.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Cursor image too large; popup hidden (z-index); autoplay error; animation plays once only; changes breaking a working game. Likely: 'it worked before!' - restore the backup.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Summative reflection (Forms): what I learned in Units 2-4, my best feature, my next goal. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: upload the final zipped game + 3-line README. Over the break: share the game with family and note one suggestion.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Starters made four features possible in one period; 70% finished all four. Students chose sounds respectfully for classmates from different backgrounds.

What will I change next time?

Give enhancements and animations two periods next year; the playtest was too short.

Students needing follow-up:

Students who did not finish (2-4 per class): extension to the first week after the break; share exemplar games.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☑3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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